



NPTEL Online Certification Course

Indian Institute of Technology Roorkee

Course Name: Power System Protection and Switchgear

Instructor: Prof. Bhaveshkumar R. Bhalja



WEEK 4: ASSIGNMENT 4

Question 1:

Additional time delay or coordination is generally not required in

- a) carrier blocking scheme.
- b) carrier tripping scheme.
- c) both (a) and (b)
- d) none of these

Answer: (b)

Question 2:

The main disadvantage of a direct underreach transfer tripping scheme is

- a) higher cost.
- b) longer operating time.
- c) maloperation due to noise or switching surges.
- d) requirement of additional time grading.

Answer: (c)

Question 3:

The frequency range of carrier signals transmitted through a microwave communication channel is typically

- a) 0.3-3 Hz
- b) 0.3-3 kHz
- c) 0.3-3 MHz
- d) 0.3-3 GHz

Answer: (d)

Question 4:

Carrier protection schemes are mainly used for

- a) High-voltage transmission lines.
- b) Low-voltage transmission lines.
- c) Distribution lines.
- d) Transformers

Answer: (a)

Question 5:

Pilot-wire relaying schemes are generally used for

- a) short transmission lines.
- b) medium transmission lines.
- c) long transmission lines.
- d) All of these

Answer: (a)

Question 6:

A carrier signal used in protection schemes can also be utilized for

- a) telemetering.

- b) supervisory control.
- c) communication.
- d) All of these

Answer: (d)

Question 7:

A 132 kV long transmission line has an impedance of $(2 + j8) \Omega$. The line is protected by MHO type of distance relay with a characteristics angle of 74° and its first zone covers 80% of the line length. The CT ratio and PT ratio are 500/1 A and 132 kV/110 V, respectively. Considering a fault resistance of 2.4Ω , determine the first zone setting of the distance relay.

- a) $K_1 = 1.27 \Omega$
- b) $K_1 = 2.73 \Omega$
- c) $K_1 = 3.27 \Omega$
- d) $K_1 = 4.73 \Omega$

Answer: (c)

Solution:

Line impedance = $2 + j8 = 8.25 \angle 75.96^\circ \Omega$

CT ratio = 500/1

PT Ratio = 132 kV/110V

Characteristic angle $(\phi) = 74^\circ$

$$Z_1 = (80\% \text{ of line impedance} + \text{fault resistance}) \times \frac{CT \text{ ratio}}{PT \text{ ratio}}$$

$$= (0.8 \times 8.25 \angle 75.96^\circ + 2.4) \times \frac{\left(\frac{500}{1}\right)}{\left(\frac{132 \times 10^3}{110}\right)} = 3.145 \angle 58^\circ \Omega$$

Hence, the first zone setting of mho relay is

$$K_1 = \frac{|Z_1|}{\cos(\theta - \phi)} = \frac{3.145}{\cos(74 - 58)} = 3.27 \Omega$$

Question 8:

A synchronism-check relay is used to

- a) ensure synchronization before closing a circuit breaker.
- b) reduce the impact on the circuit breaker.
- c) maintain system stability.
- d) None of these

Answer: (a)

Question 9:

The use of auto-reclosing improves

- a) stability of the system.
- b) continuity of service.
- c) sensitivity of protection.
- d) None of these

Answer: (b)

Question 10:

Pilot protection provides which of the following advantages?

- a) Simultaneous tripping at both ends of the line
- b) Lower cost compared to conventional protection schemes

- | | |
|----|------------------|
| c) | Both (a) and (b) |
| d) | None of these |

Answer: (a)

****END****



NPTEL Online Certification Course

Indian Institute of Technology Roorkee

Course Name: Power System Protection and Switchgear

Instructor: Prof. Bhaveshkumar R. Bhalja



WEEK 3: ASSIGNMENT 3

Question 1:

Distance protection is preferred over graded time-lag overcurrent protection for HV and EHV transmission lines because

- a) it operates faster.
- b) it is simpler.
- c) it is less expensive.
- d) All of these

Answer: (a)

Question 2:

A 3-phase, 11 kV power system has the following phase currents during an unsymmetrical fault: $I_a=100\angle 0^\circ$ A, $I_b=50\angle -120^\circ$ A, $I_c=30\angle 120^\circ$ A. The positive sequence current component is

- a) 20 A
- b) 60 A
- c) 80 A
- d) 180 A

Answer: (b)

Solution:

Positive sequence current component is,

$$I_1 = \frac{1}{3}(I_a + aI_b + a^2I_c) = \frac{1}{3}(100\angle 0 + 1\angle 120 \times 50\angle -120 + 1\angle 240 \times 30\angle 120)$$
$$I_1 = \frac{1}{3}(100 + 50 + 30) = 60 \text{ A}$$

Question 3:

The purpose of backup protection is to

- a) increase the speed of operation.
- b) increase the reach of protection.
- c) eliminate blind spots in protection.
- d) provide protection in case of failure of primary protection.

Answer: (d)

Question 4:

Which zone of a distance relay is most affected by the transient overreach phenomenon?

- a) Zone 1
- b) Zone 2
- c) Zone 3
- d) None of these

Answer: (a)

Question 5:

A short transmission line having an impedance of $(2+j5) \Omega$ is protected by a reactance relay. The CT ratio is 500/1 A and PT ratio is 132 kV/110 V. If the relay is adjusted to just operate for a dead short-circuit at the end of line section, then the set value of relay is

- a) 2.944
- b) 2.083
- c) 2.517
- d) 3.614

Answer: (b)

Solution:

$$X_{relay} = X \times \frac{CT \text{ ratio}}{PT \text{ ratio}} = 5 \times \frac{\left(\frac{500}{1}\right)}{\left(\frac{132000}{110}\right)} = 2.083$$

Question 6:

Which protective relay is commonly used for the protection of very long EHV transmission lines?

- a) Overcurrent relay with extremely inverse characteristics
- b) Percentage differential relay
- c) Mho relay
- d) Reactance relay

Answer: (c)

Question 7:

Due to current infeed from the remote end of a transmission line, the distance relay located at the local end tends to

- a) overreach.
- b) underreach.
- c) Both (a) and (b)
- d) operate normally without being affected.

Answer: (b)

Question 8:

Fig. 1 shows the single line diagram of a 400 kV three-terminal transmission line system. A three-phase fault occurs on the line section near the bus C. The magnitude and direction of currents from each bus are shown in the Fig. 1. In addition, the impedance of line section connected between bus A and T-point, between bus B and T-point and between bus C and T-point is $25\angle 80^\circ$, $15\angle 85^\circ$ and $12\angle 75^\circ$, respectively. Then, the apparent impedance seen by relay located at bus A is

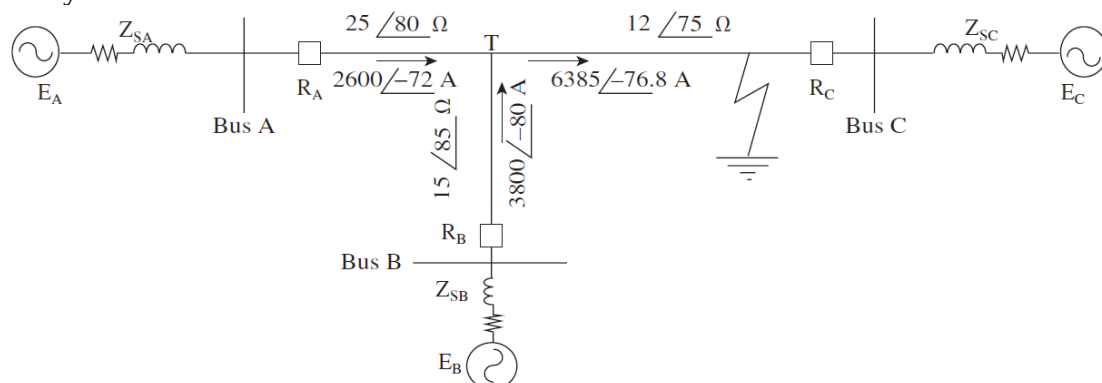


Fig. 1 Single line diagram of the three-terminal transmission line

- a) $44.3\angle 74.7^\circ \Omega$
- b) $44.3\angle 44.4^\circ \Omega$
- c) $54.3\angle 74.7^\circ \Omega$
- d) $54.3\angle 44.4^\circ \Omega$

Answer: (c)

Solution:

$$Z_A = \frac{V_A}{I_A} = \frac{12\angle 75^\circ \times 6385\angle -76.8^\circ + 25\angle 80^\circ \times 2600\angle -72^\circ}{2600\angle -72^\circ} = 54.3\angle 74.7^\circ \Omega$$

Question 9:

Overreaching of a distance relay caused by the decaying DC component can be minimized by using

- a) electromechanical relay.
- b) solid-state relay.
- c) digital relay.
- d) None of these

Answer: (c)

Question 10:

A 132 kV transmission line has an impedance of $(4 + j16) \Omega$. This line is protected by mho type distance relay with CT ratio = 1000/1 A and PT ratio = 132 kV/110 V. The zone-1 of the reactance relay covers 80% of the line length, zone-2 covers 150% of the line length. Then, the first and second zone setting of reactance relay are

- a) 20.67Ω and 10Ω , respectively
- b) 10.67Ω and 20Ω , respectively
- c) 10Ω and 20.67Ω , respectively
- d) 20Ω and 10.67Ω , respectively

Answer: (b)

Solution:

$$Z_{line} = 4 + j16 \Omega, \quad X_{line} = 16 \Omega$$

Zone-1 setting (80% of line)

$$X_{Zone-1} = 0.8 \times X_{line} \times \frac{CT \text{ ratio}}{PT \text{ ratio}} = 0.8 \times 16 \times \frac{\left(\frac{1000}{1}\right)}{\left(\frac{132000}{110}\right)} = 10.67 \Omega$$

Zone-2 setting (150% of line)

$$X_{Zone-2} = 1.5 \times X_{line} \times \frac{CT \text{ ratio}}{PT \text{ ratio}} = 1.5 \times 16 \times \frac{\left(\frac{1000}{1}\right)}{\left(\frac{132000}{110}\right)} = 20 \Omega$$

*****END*****



WEEK 2: ASSIGNMENT 2

Question 1:

The main drawback of a definite time delay relay is that

- a) it takes a long time to clear close-in faults.
- b) it is unable to clear far-end faults.
- c) both (a) and (b).
- d) None of these

Answer: (a)

Question 2:

The limitations of electromechanical relays include

- a) presence of moving parts leading to friction issues.
- b) low torque production.
- c) high burden and power consumption for auxiliary mechanisms.
- d) All of these

Answer: (d)

Question 3:

A 60 MVA, 33 kV/66 kV, star/delta power transformer is to be protected by differential protection. The continuous current carrying capacity to restraining coils of the differential relay should not exceed 5 A. CT ratio is 3000/5 on the 33 kV side. What would be the current fed into the pilot wires of the CT secondary in the 33 kV side?

- a) 2.13 A
- b) 3.03 A
- c) 4.23 A
- d) 4.85 A

Answer: (b)

Solution:

On 33 kV side,

$$I_F = \frac{60000}{\sqrt{3} \times 33} = 1049.7 \text{ A}$$

CT ratio: 3000/5

$$I_{F(\text{secondary})} = \frac{1049.7}{\left(\frac{3000}{5}\right)} = 1.75 \text{ A}$$

Since, 33 kV is star connected and CT secondary will be connected in Δ . Hence, current fed into pilot wires is:

$$= \sqrt{3} \times 1.75 = 3.03 \text{ A}$$

Question 4:

In continuation of question number 3, what could be the possible CT ratio on 66 kV side?

- a) 200/5
- b) 500/5
- c) 1000/5

d) 1500/5

Answer: (c)

Solution:

On 33 kV side,

$$I_F = \frac{60000}{\sqrt{3} \times 66} = 524.85 \text{ A}$$

Current in secondary of CT's = Current in pilot wires

Since 66 kV side CT secondary are connected in star with current = 3.03 A

$$CT \text{ ratio} = \frac{524.85}{3.03} = 173.22$$

Take CT ratio = 200, Secondary current = 5 A,

Hence, the CT ratio on 66 kV side = 1000/5

Question 5:

Which is the closest relay characteristic to fuse/MCCB?

- a) Normal inverse
- b) Very inverse
- c) Extremely inverse
- d) All of these

Answer: (c)

Question 6:

In an electromagnetic type overcurrent relay having an inverse definite minimum time (IDMT) characteristic, the minimum-time feature is achieved by

- a) saturation of the magnetic circuit.
- b) proper mechanical design.
- c) an appropriate time-delay element.
- d) electromagnetic damping.

Answer: (a)

Question 7:

Fig. 1 shows the single line diagram of a portion of a radial distribution system. The plug setting (PS) of $R_3 = 75\%$ of CT secondary. The TDS of $R_3 = 0.1$. The normal range of PS is 50-200% of 1 A in steps of 25%, whereas the TDS setting range is 0.1 to 1 s in steps of 0.05. What would be the PSs of the relays R_1 and R_2 ?

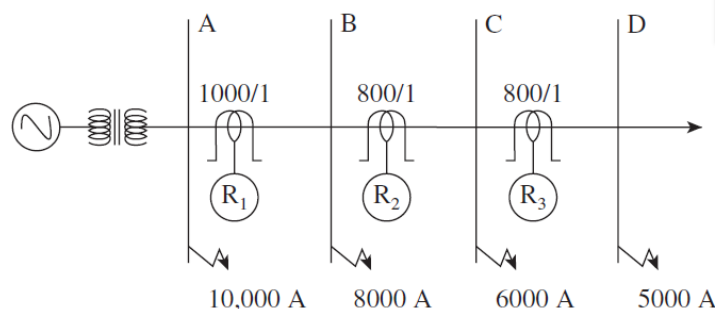


Fig.1 Single line diagram of radial distribution system

- a) 75% and 100% of relay rating, respectively.
- b) 100% of relay rating for both.
- c) 100% and 75% of relay rating, respectively.
- d) 75% of relay rating for both

Answer: (b)

Solution:

Plug setting (PS) of $R_3 = 75\%$

CT ratio (R_1) = 1000/1, CT ratio (R_2) = 800/1, CT ratio (R_3) = 800/1

Pickup current of R_3 in secondary = $0.75 \times 1 = 0.75$ A

Pickup current of R_3 in primary = $0.75 \times 800 = 600$ A

For proper coordination:

R_2 must have a pickup current greater than R_3

R_1 must have a pickup current greater than R_2

If PS of R_2 is set at 75% then the pickup current of R_2 in primary = $0.75 \times 800 = 600$ A

Since it is the same as the pickup current of R_3 , which is not desirable for coordination.

So, at the next higher PS of 100%, the pickup current of R_2 in primary = $1 \times 800 = 800$ A

If PS of R_1 is set at 75% then the pickup current of R_1 in primary = $0.75 \times 1000 = 750$ A

Since it is less than the pickup current of R_2 , which is not desirable for coordination.

So, at the next higher PS of 100%, the pickup current of R_1 in primary = $1 \times 1000 = 1000$ A

Question 8:

In continuation of question number 7, assuming the coordination time interval between two successive relays as 0.3 s, what is the Time Dial Setting (TDS) of the relay R_2 ?

- a) 0.10
- b) 0.15
- c) 0.20
- d) 0.25

Answer: (c)

Solution:

$$MP(R_3) = \frac{\text{Fault current}}{PS \times CT \text{ ratio}} = \frac{5000}{0.75 \times \left(\frac{800}{1}\right)} = 8.33$$

$$t_{op}(R_3) = \frac{0.14}{MP^{0.02} - 1} \times TDS \text{ of } R_3 = \frac{0.14}{8.33^{0.02} - 1} \times 0.1 = 0.323$$
$$t_{op}(R_2) = 0.323 + 0.3 = 0.623$$

For the same fault current,

$$MP(R_2) = \frac{\text{Fault current}}{PS \times CT \text{ ratio}} = \frac{5000}{1 \times \left(\frac{800}{1}\right)} = 6.25$$

$$t_{op}(R_2) = \frac{0.14}{MP^{0.02} - 1} \times TDS \text{ of } R_2$$

$$0.623 = \frac{0.14}{6.25^{0.02} - 1} \times TDS \text{ of } R_2$$
$$TDS \text{ of } R_2 = 0.166$$

Nearest higher setting is 0.2

Therefore, $TDS \text{ of } R_2 = 0.2$

Question 9:

A thermal relay generally has

- a) delayed trip action
- b) instantaneous trip action

- c) both delayed and instantaneous trip action
- d) None of these

Answer: (a)

Question 10:

The transient overreach phenomenon in an overcurrent relay increases when the decay of the DC component of fault current is

- a) Constant
- b) Fast
- c) Slow
- d) None of these

Answer: (c)

****END****



NPTEL Online Certification Course

Indian Institute of Technology Roorkee

Course Name: Power System Protection and Switchgear

Instructor: Prof. Bhaveshkumar R. Bhalja



WEEK 1: ASSIGNMENT 1

Question 1:

Unit protection is based on the concept of

- a) Absolute selectivity
- b) Relative selectivity
- c) Both (a) and (b)
- d) None of these

Answer: (a)

Question 2:

Find the number of 11 kV suspension insulator discs required for a three phase 220 kV power transmission line, with a safety factor of 1.5?

- a) 12
- b) 18
- c) 20
- d) 30

Answer: (b)

Solution:

Line voltage (V_L) = 220 kV

Phase voltage (V_P) = $220/\sqrt{3}$ kV

Voltage rating of each insulator disc (V_r) = 11 kV

Safety factor (S_f) = 1.5

$$\text{Required number of 11 kV insulator discs} = \frac{V_P}{V_r} \times S_f = \frac{220/\sqrt{3}}{11} \times 1.5 = 17.32 \approx 18$$

Question 3:

What is the main drawback of electromechanical and static relays?

- a) Both are very costly.
- b) They are not rugged in nature.
- c) Lack of continuous monitoring of their operational integrity.
- d) None of these

Answer: (c)

Question 4:

A thermal relay is used for protection against

- a) Overload
- b) Overvoltage
- c) Short-circuit
- d) All the above

Answer: (a)

Question 5:

Intelligent Electronic Devices (IEDs) are useful for

- a) Protection

- b) Measurement and storing
- c) Control
- d) All of these

Answer: (d)

Question 6:

The operating time of a modern digital relay is typically of the order of

- a) 0 – 10 ms
- b) 10 – 30 ms
- c) 50 – 100 ms
- d) 100 – 200 ms

Answer: (b)

Question 7:

The VA rating of a current transformer (CT) is determined by considering

- a) total burden connected to the CT secondary circuit.
- b) resistance of the relay only.
- c) reactance of the relay only.
- d) secondary current rating only.

Answer: (a)

Question 8:

The least number of faults is generally reported for

- a) transmission line
- b) cables
- c) switchgear
- d) transformers

Answer: (b)

Question 9:

The apparent power (VA) required to operate a relay is known as

- a) stability of the relay
- b) tripping time of the relay
- c) sensitivity of the relay
- d) None of these

Answer: (c)

Question 10:

A protective relay is used to

- (i) provide additional safety by limiting arcing current during circuit breaker operation.
- (ii) close its contacts when the actuating quantity reaches a predetermined value.

- a) Only (i)
- b) Only (ii)
- c) Both (i) and (ii)
- d) None of these

Answer: (b)

*****END*****